



Infrastructure  
Canada

Canada

# THE CLIMATE LENS

Investing in Canada  
Infrastructure Program  
General Guidance v 2.1

Aussi disponible en français sous le titre : L'optique des changements climatiques : Programme d'infrastructure Investir dans le Canada - Lignes directrices générales.

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## Disclaimer

This guidance document is for those applicants applying for funding under **the Investing in Canada Infrastructure Program (ICIP)** of Infrastructure Canada (INFC) and who are obligated to meet the Climate Lens requirement. Hence, the climate change impacts requested in this document are a direct reflection of the parameters of the Climate Lens and INFC funding requirements only.

This document is intended to be a learning tool for project developers and to introduce climate change considerations into project designs in the context of the Canadian environment. This guidance document is evergreen – meaning it will be periodically updated to remain aligned with advancing assessment methodologies. For the most recent version of the guidance, please consult the Infrastructure Canada website.

## Revision History

| Revision No. | Issue Date     | Revision Description                                                                                                                                                                                   |
|--------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version 1.0  | June 2018      | First version implemented.                                                                                                                                                                             |
| Version 1.2  | September 2019 | General revisions after internal review.                                                                                                                                                               |
| Version 2.0  | March 2021     | New format and approach introduced. Single form with streamlined GHG and resilience requirements.                                                                                                      |
| Version 2.1  | January 2023   | Revisions based on feedback from Version 2.0. Climate Lens Guidance now two documents:<br><a href="#">DMAF Climate Lens Guidance v 2.1</a><br><a href="#">ICIP Climate Lens General Guidance v 2.1</a> |

# Table of Contents

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|                                                                                 |    |
|---------------------------------------------------------------------------------|----|
| 1.0 Introduction .....                                                          | 5  |
| 1.1 What is the Climate Lens? .....                                             | 5  |
| 1.2 Applicable Programs .....                                                   | 5  |
| 1.3 Cost Eligibility .....                                                      | 6  |
| 1.4 Responsible Party .....                                                     | 7  |
| 1.5 Infrastructure Canada Review of the Climate Lens .....                      | 7  |
| 2.0 Climate Lens Guidance .....                                                 | 8  |
| 2.1 Step by Step Instructions .....                                             | 9  |
| 2.2 Submission of the Climate Lens .....                                        | 18 |
| ANNEX A – Climate Lens Form .....                                               | 19 |
| ANNEX B – Average P/T Grid Electricity Emission Intensities (tonnes/MWh)* ..... | 22 |
| ANNEX C – Project-Specific GHG Baseline & Project Scenarios .....               | 24 |

# 1.0 Introduction

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The purpose of this document is to provide guidance to those who need to undertake a Climate Lens. The objectives of this guidance are to:

1. Explain the purpose of the Climate Lens and which projects are subject to the requirement;
2. Provide information on when and how to submit a completed Climate Lens to Infrastructure Canada;
3. Provide step by step instructions on how to complete the various sections of the Climate Lens.

## 1.1 What is the Climate Lens?

The Government of Canada is committed to meeting its 2030 greenhouse gas reduction target, establishing a cleaner, more competitive and resilient economy and getting Canada to net-zero emissions by 2050. The Climate Lens is a key tool for assessing the climate impacts of infrastructure from both a greenhouse gas and resilience perspective. The tool also encourages applicants to consider how their projects can reduce GHG emissions and increase resilience to climate change, which benefits their communities and creates jobs.

The Climate Lens is a project-level requirement applicable to Infrastructure Canada's Investing in Canada Infrastructure Program (ICIP) and Disaster Mitigation and Adaptation Fund (DMAF). The main goal of the Climate Lens is to raise awareness of climate change risks and impacts associated with projects and encourage improved choices by project planners, designers and decision-makers. The Climate Lens also supports Infrastructure Canada in measuring its progress towards meeting its climate goals.

The Climate Lens has two key sections: **GHG Emissions and Mitigation**, which looks at the anticipated greenhouse gas (GHG) emissions impact of an infrastructure project; and **Climate Resiliency**, which examines the risk and resilience of the project to climate change related disruptions or impacts.

## 1.2 Applicable Programs

The table below identifies the programs, streams and sub-streams to which the Climate Lens applies and lists the project value thresholds at which the Climate Lens is required.

Projects submitting under the Green Infrastructure – Climate Change Mitigation sub-stream of ICIP are expected to demonstrate an overall reduction in emissions compared to the business-as-usual (BAU) scenario. Projects submitting under the Green-



Infrastructure – Adaptation and Resilience sub-stream are expected to demonstrate an overall increase of resiliency to climate impacts.

**Table 1.** Thresholds for Climate Lens Requirement under ICIP

| ICIP Streams                                                                                                                                    | Climate Lens                                           | When to submit                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------|
| <b>Green Infrastructure – Climate Change Mitigation</b> sub-stream                                                                              | All projects                                           | Climate Lens due at time of application. |
| <b>Green Infrastructure – Adaptation, Resilience and Disaster Mitigation</b> sub-stream                                                         | All projects                                           | Climate Lens due at time of application. |
| <b>Other streams and Sub-streams</b> (Environmental Quality, Public Transit, Community, Culture and Recreation, Rural and Northern Communities) | If total eligible project costs are greater than \$10M | Climate Lens due at time of application. |

Note 1. Applicants from Quebec required to complete a Climate Lens under ICIP are to follow the guidelines for submission as outlined in their Integrated Bilateral Agreement.

## 1.3 Cost Eligibility

Infrastructure Canada strongly encourages applicants to perform analyses informed by best practices regarding GHG mitigation and climate risks and resilience in their projects. For this reason, the costs of undertaking the Climate Lens will be deemed eligible for cost-sharing for all projects approved for federal funding. This includes all associated costs incurred such as preparation and supporting analysis in keeping with best practices. Any costs incurred to undertake GHG Mitigation and/or Climate Resilience assessments under previous versions of the Climate Lens guidance remain eligible for reimbursement, except where own-force labour is used.

Should applicants to the ICIP with projects beneath the \$10M threshold wish to complete a Climate Lens on an optional basis, these costs would be eligible for cost-sharing if the project is approved for federal funding, as long as the Climate Lens conforms to the requirements outlined here and is submitted to Infrastructure Canada at the time of application.

Since costs are only eligible for cost-sharing for federally approved projects, municipalities, Indigenous communities, and other applicants are encouraged to

engage regularly with the relevant province or territory to determine their project's likelihood of prioritization before incurring costs related to undertaking a Climate Lens.

## 1.4 Responsible Party

It is the applicant's responsibility to ensure that the Climate Lens is completed by someone with appropriate qualifications and knowledge of the project, as determined by the applicant. This could be the applicant, the applicant's design consultant, or another consulting body. If the applicant determines that the needed qualifications are not available on the project team, Infrastructure Canada recommends engaging a qualified professional, such as a professional engineer, GHG accounting professional, or registered professional planner. Infrastructure Canada is able to provide further advice and recommendations on selecting an appropriate professional at the applicant's request.

## 1.5 Infrastructure Canada Review of the Climate Lens

Infrastructure Canada will review each Climate Lens assessment and may follow up on the results of the Climate Lens to confirm the information submitted or to request further detail. Climate Lens assessments for projects in the Climate Change Mitigation and Adaptation, Resilience and Disaster Mitigation sub-streams of the Investing in Canada Infrastructure Program will be assessed appropriately to ensure program requirements have been met.

**The inclusion by applicants of supporting information, such as calculations, assumptions justification and references – either in the Climate Lens form itself or as an attachment – will help minimize the need for follow-ups and facilitate a timely review by Infrastructure Canada.**

Applicants are encouraged to contact Infrastructure Canada's Climate Lens Policy Team at: [climatelens-optiquedeschangementsclimatiques@infcc.gc.ca](mailto:climatelens-optiquedeschangementsclimatiques@infcc.gc.ca) for further assistance and/or links to other resources as necessary.



## 2.0 Climate Lens Guidance

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The Climate Lens is divided into four sections:

1. **Project Overview** provides administrative information and a description of the project;
2. **GHG Emissions & Mitigation** looks at the anticipated greenhouse gas (GHG) emissions impact of the infrastructure project;
3. **Climate Resiliency** examines the risk and resilience of the project to a climate change related disruption or impact;
4. **Climate Objectives** provides additional and supporting information to the overall climate objectives of the project and the applicant.

The following section provides guidance on how to complete each section of the Climate Lens. Additional guidance on acceptable approaches to quantifying GHG emissions reductions and the identification of climate risks is provided on Infrastructure Canada's website.

Applicants may still submit Climate Lens assessments according to the previous versions of the Climate Lens Guidance (version 1.0-2.0), if so desired.

Note that ICIP projects from Quebec are to continue to follow the provincial guidelines to GHG quantification as outlined in the Canada-Quebec Integrated Bilateral Agreement.

### As a reminder...

- Projects under all ICIP streams (including those in the Green Climate Change Mitigation and Green Adaptation and Resilience stream under \$10 million) are required to complete the Climate Lens form.
- Please ensure to complete each section of the form and provide an answer to each question.
- Please ensure all supporting information has been included in the form or attached as a separate document. All data, calculations and assumptions used to answer the questions in the Climate Lens should be provided to help assist in the review and minimize the need for follow-ups.
- Any changes to the project or the project timeline that occur after the submission of the Climate Lens must be reported to Infrastructure Canada as part of the regular progress reporting requirements and Infrastructure Canada will advise if a revised Climate Lens will need to be submitted.

## 2.1 Step by Step Instructions

Applicants are to follow the guidance provided in this section to complete the Climate Lens Form found in ANNEX A.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.0 Project Overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>1.1 Project Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Title of your project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>1.2 Ultimate Recipient</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Full legal identification of the primary entity that is undertaking the project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>1.3 Project Description: location, activities and timeline</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>Present a <b>brief description</b> of the project including <b>location, all major activities and its timeline</b> occurring on the site.</p> <p><u>Location:</u><br/>Provide the address and/or GPS coordinates of the project. Provide a map to indicate the location of the project site, illustrating the boundary of the project and ensuring all relevant components of the project are included.</p> <ul style="list-style-type: none"><li>• Identify whether the project site is currently vegetated, a wetland or a peat bog, on permafrost or is considered a brownfield.</li><li>• Identify whether the project site is accessible by public transportation or active transportation (bicycle/walking).</li></ul> <p><u>Description of Activities:</u><br/>The project description lays out the foundation for the types of activities that may be subject to climate risk, or which may release or sequester GHGs from the project. These activities must be quantified in the Climate Lens.</p> <ul style="list-style-type: none"><li>• Describe the product or service provided by the project, including a description of any project-specific technologies that will be implemented.</li><li>• Describe the services or output that will be provided and all major activities that will occur on the project site.</li><li>• The type(s) of technologies that will be used can also be described here.</li></ul> <p><u>Timeline:</u><br/>Detail the project timeline outlining the timing of operational activities. Specifically, the following estimated dates are required:</p> <ul style="list-style-type: none"><li>• Construction start and end dates;</li><li>• Operational start and end dates;</li><li>• Dates of any major maintenance/repairs/refurbishments expected;</li><li>• Expected lifetime of the project.</li></ul> <p>Identify any risks that could substantially affect the project's operational timelines.</p> |

## 2.0 GHG Emissions & Mitigation

### 2.1 Is your project a GHG mitigation project OR are you implementing any GHG mitigation measures or best practices in the design of your project?

If yes, describe how your project reduces GHG emissions and/or describe the project components that will result in a decrease in GHG emissions (see below for examples).

If no, describe examples of project elements that were considered and explain why they were not implemented (e.g., solar panels were too expensive for the project budget). Then skip to section 3.0 Climate Resilience.

#### **GHG Mitigation Projects:**

- Renewable energy generation projects;
- Interties;
- Electrification of industrial facilities (e.g., natural gas processing);
- Large transportation projects (LRT/BRT);
- Vehicles using clean fuels or zero-emission vehicles;
- Waste & wastewater – e.g., flaring, biogas utilization;
- Retrofits ex. installing energy efficient HVAC systems, other equipment, lighting, smart monitoring or removal of high GWP refrigerants.

#### **Projects implementing measures or best practices may include:**

- Using clean technologies such as wind, solar, or geothermal energy or energy storage such as batteries, whether alone or integrated into other assets (e.g., a building);
- New buildings that: are Green Design (LEED) certified; with Energy Star ratings above 75; and/or use lower-carbon forms of heating and cooling such as heat pumps;
- Implementing any new equipment to decrease process emissions (e.g., biofiltration in a wastewater treatment facility).

### 2.2 If your project is a GHG mitigation project or you are implementing GHG mitigation measures, what are the annual GHG emission reductions (tonnes CO<sub>2</sub>e/year) expected in 2030 from the operation of the project?

Please provide an estimate of the operational GHG reductions expected in 2030 and for each year (if expected to be different) over the lifetime of the project. These values are best shown in a table format. Note that most projects, which expect to have consistent operations over the course of their lifetime, will likely be able to report very similar annual reductions over the lifetime of the project. Larger projects such as transit or renewable energy projects or projects being implemented in phases could see larger variations in their annual emission reductions.

**Supporting information, such as the description of the baseline, all main sources/sinks, emission factors, calculations and assumptions, will in most cases be required by Infrastructure Canada to conduct its review of the Climate Lens and should be included.**

The standard equation to calculate GHG emission reductions is the following:

$$\text{Baseline Emissions} - \text{Project Emissions} = \text{GHG Emission Reductions}$$

Where:

The **Baseline** is the “business as usual” (BAU) scenario or hypothetical reference case against which the GHG performance of the project is measured.

The **Project** represents the new project applying for funds under Infrastructure Canada's funding programs.

Please consult Annex C for further information on pre-selected baseline and project scenarios that should be used for various project types under the Climate Lens. Annex B provides provincial and territorial average grid electricity emission intensities to be used where applicable (when calculating GHG emissions from electricity consumed).

As an additional tool to assist applicants, Infrastructure Canada will provide sector-specific guidance for calculating GHG emission reductions for certain project types. This guidance will be made available in phases on Infrastructure Canada's website. If your project type is not currently supported by Infrastructure Canada's available sector-specific guidance, you may consider referring to the references noted in Section 2.3 for support in completing GHG quantification.

For those applicants unable to provide a reliable estimate of the GHG emissions reductions at time of application, please explain why (e.g., methodology not yet established, missing project details due to early stages of development) and the estimated date at which the information will be submitted. **An estimate of GHG emissions reductions will be required before first claim.** Those projects under the Green Infrastructure – Climate Change Mitigation Stream will be required to provide an initial estimate of GHG reductions at time of application, and may be requested to provide an updated, more detailed estimate prior to first claim, following Infrastructure Canada's review.

Applicants can refer to the Climate Lens website for additional resources on GHG mitigation or contact the Climate Lens Team at: [climatelens-optiquedeschangementsclimatiques@infc.gc.ca](mailto:climatelens-optiquedeschangementsclimatiques@infc.gc.ca) for additional support.

### **2.3 Which international GHG quantification standards, GHG guidance or other supporting resources were consulted to understand the GHG impact of the project?**

Please list the resources you consulted to identify the GHG emissions and/or reductions resulting from your project. Other supporting information could include Impact Assessment studies, GHG or climate risk assessments, energy audits, benchmark studies, certification applications, etc.

There are a variety of resources that can be consulted to help quantify the GHG emissions/reductions from your project such as the following:

- o [ISO 14064:2](#)

- [World Resources Institute \(WRI\) GHG Protocol](#)
- [Federation of Canadian Municipalities: Guidebook on Quantifying GHG Reductions at the Project Level](#)
- [ICLEI/C40 Cities/WRI GHG Protocol Global Protocol for Community Scale GHG Emission Inventories](#)
- [US Community Protocol for Accounting and Reporting of GHG Emissions](#)
- [Alberta Offset Protocols](#)
- [Climate Action Reserve Protocols](#)
- [Intergovernmental Panel on Climate Change \(IPCC\) Guidelines for GHG Inventories](#)
- [UNFCCC Clean Development Mechanism Methodologies](#)

### 3.0 Climate Resiliency

**3.1 Identify all current and projected climate-related hazards given the project's location such as flooding, wildfire risk, permafrost thaw or coastal erosion. Assess high or medium risks (in likelihood and severity) to the project and the services it is to provide over its lifespan.**

Provide a list and brief description of all the current and future climate-influenced hazards facing the project over its lifetime. Climate-influenced hazards may include: increasing temperatures, extreme heat, drought, wildfires, freeze-thaws, increased snow loads, increasing precipitation, ice, storms, earthquakes, etc.

You may wish to consult municipal or regional climate adaptation reports to identify the climate-influenced hazards that are relevant to your project location. For each hazard, it is recommended to look at both the current and future climate data projections for the lifespan of the project. (E.g., if your project will be complete in 2024 and has an expected lifespan of 30 years, you will want to look at current climate data as well as the 2050 RCP-8.5 climate projections).

The climate projections can be provided in a chart or written format, describing how these parameters are projected to change against the historical baseline. (E.g., according to The City's Climate Risk Report, precipitation is expected to increase by 11% over current historical baseline by 2080 under a high climate change scenario).

You may wish to consult the Canadian Centre for Climate Services for support.

Example Chart:

| Climate influenced hazard     | Historic value (1976-2005) | Short- term forecast (2020-2050)/ percent change | Long term forecast (2050-2080)/ percent change | Data source     |
|-------------------------------|----------------------------|--------------------------------------------------|------------------------------------------------|-----------------|
| Number of very hot days (+30) | 1 day annually             | 4.9 annually (+3.9% )                            | 16.3 days annually (+15.2% )                   | Climateatlas.ca |
| Etc.                          |                            |                                                  |                                                |                 |

In order to understand the risks associated with the natural hazards identified above the various components of the project should be assessed against each hazard. The likelihood of the hazard impacting the project, and the consequence or level of severity of that impact, should also be analyzed. Using both the likelihood and severity of a hazard will help identify those aspects of a project that are at highest risk to climate impacts (e.g., High Risk = Very High likelihood + Very high consequence).

Please follow Steps 1 to 4 to assess the risks of the natural hazard to your project. Please include a written description of which project components will be subject to the highest risks and why. All risks can additionally be listed in the Table provided under Step 4.

For example:

- Wildfires will present a risk to a community centre project due to its location near a forested area that is experiencing increased occurrence of drought and increasing temperatures; or
- Projected increases in rainfall will present a high risk to bus shelters as it can lead to flash flooding that can cause damage to the physical infrastructure, obstruct access for bus users, cause disruptions to maintenance work, and result in delays of services.

**Step 1:** Assess the severity or consequence of the hazard's impact on the project.

- Consequence refers to the impact of the hazard on the project. Note a hazard can lead to a range of consequences. A consequence can be certain or uncertain and can have both positive or negative effects on the project's objectives.
- Determine the level of consequence from very-low to very high.

**Step 2:** Assess the likelihood of the hazard's impact on the project.

- Likelihood can be understood as the chance of something occurring or the chance of a defined climate hazard over a given time horizon. Applicants should feel free to use their own discretion when determining the likelihood risk for each hazard, but are encouraged to consult additional climate data relevant to the hazards identified. Additional resources are listed in Section 3.3.
- Determine the level of likelihood from very low to very high.

**Step 3:** Identify the number and level of risks of each hazard on the project.

- Using the matrix below in table 3.0, assess the risk level for each interaction by finding where the consequence risk and likelihood risk meet.
- For each hazard identified record whether the hazard poses a low, medium, or high risk to the project.



Table 3.0

|                     |           |                 |                 |             |             |             |
|---------------------|-----------|-----------------|-----------------|-------------|-------------|-------------|
| <b>Consequences</b> | Very High | Medium Risk     | High Risk       | High Risk   | High Risk   | High Risk   |
|                     | High      | Low Risk        | Medium Risk     | High Risk   | High Risk   | High Risk   |
|                     | Moderate  | Low Risk        | Low Risk        | Medium Risk | High Risk   | High Risk   |
|                     | Low       | Negligible Risk | Low Risk        | Low Risk    | Medium Risk | Medium Risk |
|                     | Very Low  | Negligible Risk | Negligible Risk | Low Risk    | Low Risk    | Low Risk    |
|                     |           | Very Low        | Low             | Moderate    | High        | Very High   |
| <b>Likelihood</b>   |           |                 |                 |             |             |             |

#### Step 4: Record Risks

If helpful, applicants may complete the following template to identify to risks of each hazards.

| <b>Climate-Influenced Hazard</b> | <b>Consequence Risk</b> | <b>Justification of Consequence Risk</b>                                  | <b>Likelihood Risk</b> | <b>Justification of Likelihood Risk</b>                                      | <b>Overall Risk</b> |
|----------------------------------|-------------------------|---------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------|---------------------|
| Flooding                         | High                    | Project could be damaged by flooding if 1-in-100 year flood were to occur | Moderate               | Level of flood required to damage or impact the project is rare in frequency | Medium risk         |

Note, INFC suggests the following definition based on the definition by the *Public Infrastructure Engineering Vulnerability Committee (PIEVC) Protocol* (see [www.pievc.ca](http://www.pievc.ca)):

- Negligible Risks may not require further consideration
- Low Risks may require action
- Medium Risks require action
- High Risks require action

Please list any methodology that was used to assess future climate risks such as ISO 31000, PIEVC or the PIEVC High Level Screening Guide. Applicants can refer to the Climate Lens website for additional support / methodologies for assessing climate risk.

If no climate risks were identified, please justify why (e.g., the climate risks examined only pose a minimal risk). Infrastructure Canada may follow up if known potential climate hazards are missing or the analysis does not align (e.g., a community centre on a coastline is not considering risk of sea level rise).

### 3.2 Describe all of the risk mitigation measures that will be taken to improve the climate resiliency of your project.

Identify measures for all of the medium and high risks identified in 3.1. If no measures are being taken to improve the climate resiliency of your project, please describe why. If your project is a protective infrastructure, please mention that.

- Please identify and describe risk treatment or adaptation measures for all medium and high risks in order to reduce unacceptable risks to acceptable levels.
  - Your response should address all risks identified in 3.1, and describe how they will address the specific climate risks to the project.
- Examples include building a seawall or restoring wetlands to address flooding; providing firebreaks to decrease severity of wildfires; installing flooding sensors in elevators; or elevating electrical and HVAC systems to minimize flood risk. This can include considering nature-based solutions.
- Please list any resilience standards, guidance, or tools that were consulted. For example: CSA S900.1:18 Climate change adaptation for wastewater treatment plants; CSA PLUS 4011-19 Technical guide: Infrastructure in permafrost: A guideline for climate change adaptation; and the National guide for wildland-urban-interface fires. For guidance and standards that incorporate climate resilience visit: [Infrastructure Canada – Codes, Standards and Guidance for Climate Resilience](#)
- If climate risk reduction measures were identified but not implemented, please justify why not.
- Infrastructure Canada may follow up and request more information on why resiliency measures have not been taken if the justification is not clear, or if resilience measures for potential climate risks identified in 3.1 are missing.
- Applicants can refer to the Climate Lens website for additional resources on climate adaptation.

### 3.3 Please list all of the climate change data and tools that were used to determine the risks to your project.

List the climate data and tools, such as future climate projections available through the Canadian Centre for Climate Services, that were consulted to assess any current and future climate risks to your project.

- Examples include:
  - Pacific Climate Impacts Consortium: [Design Value Explorer](#)
  - ClimateData.ca: <https://climatedata.ca/>
  - Canadian Centre for Climate Services: <https://www.canada.ca/en/environment-climate-change/services/climate-change/canadian-centre-climate-services.html>
  - Climate Atlas of Canada: <https://climateatlas.ca/home-page>
  - Platform for the Analysis and Visualization of Climate Science: <https://ouranosinc.github.io/pavics-sdi/>

- Applicants can refer to the Climate Lens website for additional support and resources.

## 4.0 Climate Objectives

### 4.1 Does your community / municipality have a Climate Action Plan and if yes, does your project align with this plan?

- Please indicate the specific community/municipal climate action plan.
- This can be a stand-alone climate action plan or integrated into a broader Strategic Plan.
- Examples include the City of Kelowna, British Columbia's [Community Climate Action Plan 2018-2023](#) and the Halifax Regional Municipality, Nova Scotia's [HaliFACT: Acting on Climate Together](#).
- Please describe how the project (or measures being considered) fits into the climate action plan, and how this will contribute to a more sustainable future for your community.

## 5.0 Attestation

I, the undersigned, as authorized by my organization, confirm the statements above are true and accurate, and attest that:

- opportunities to quantify and minimize GHG emissions during the construction and operation of the project will be considered in the planning, design and development/ implementation of this project to the extent possible;
- and, climate change risks and adaptation and resiliency measures will be considered in the siting/location, design/build, and planned operation and maintenance of this project to the extent possible and reflecting the project's cost, criticality and vulnerability.

Infrastructure Canada may follow up on the results of the Climate Lens to confirm the required information or to request further detail. Consequently, applicants should retain all the information used to complete the Climate Lens.

Signature of person responsible for completing the Climate Lens:

X

Date:

Name:  
Position:  
Organization:  
Address:  
Contact Number:  
Email:

## 2.2 Submission of the Climate Lens

For projects under the Investing in Canada Infrastructure Program, the Climate Lens is to be completed and submitted in its entirety at the time of application, unless otherwise indicated by Infrastructure Canada.

**Please ensure all information and supporting documentation is included or attached with the Climate Lens form at time of submission.**

Applicants indicating the potential for GHG emissions reductions but who are unable to provide an accurate estimate in Section 2.3 may submit a rationale for being unable to do so at time of application. If Infrastructure Canada accepts the rationale and the project is approved, the applicant will be asked to provide an estimate of GHG emissions reductions before first claim. Those projects under the Green Infrastructure – Climate Change Mitigation Stream are required to provide an initial estimate at time of application, and may be requested to provide an updated, more detailed estimate prior to first claim, depending on the results of Infrastructure Canada's review of the assessment.

Projects are to be submitted to Infrastructure Canada via the Infrastructure Recipient Information System (IRIS) digital portal (or equivalent), unless otherwise stated in program guidelines.

If the project requires changes to its components, process or timeline, a revised Climate Lens may need to be submitted. Only project changes that may impact the GHG emissions estimate significantly may trigger the need for a re-assessment. Infrastructure Canada will review the changes and will contact the applicant if a revised Climate Lens is required.

Infrastructure Canada will make the Climate Lens and all supporting guidance available via provincial and territorial contacts as well as on the Infrastructure Canada website.

The applicant, via the province or territory, is responsible for completing the Climate Lens and providing the completed form to Infrastructure Canada. It is the applicant's responsibility to meet any request for further information from Infrastructure Canada.

# ANNEX A – Climate Lens Form

## ICIP Climate Lens Version 2.1



Infrastructure  
Canada

### 1.0 Project Overview

#### 1.1 Project Title

#### 1.2 Ultimate Recipient

#### 1.3 Project Description: Location, activities, timeline

**Location :**

**Activities :**

**Timeline :**

| Phase                         | Start Date- DD/MM/YYYY | End Date- DD/MM/YYYY |
|-------------------------------|------------------------|----------------------|
| Construction                  |                        |                      |
| Operation                     |                        |                      |
| Any major maintenance/repairs |                        |                      |

Expected Lifetime:

Description of any maintenance, repairs, refurbishments expected:

### 2.0 GHG Mitigation Assessment

**2.1 Is your project a GHG mitigation project OR are you implementing any GHG mitigation measures or best practices, such as clean technologies, renewable energy or LEED standards in the design of your project?**

**2.2 If your project is a GHG mitigation project or you are implementing GHG mitigation measures, what are the annual GHG emission reductions (tonnes CO<sub>2</sub>e/year) expected in 2030 from the operation of the project?**

**2.3 Which international GHG quantification standards or GHG guidance was consulted to understand the GHG impact of the project?**

### **3.0 Climate Resiliency**

**3.1 Identify all current and projected climate-related hazards given the project's location such as flooding, wildfire risk, permafrost thaw or coastal erosion. Assess high or medium risks (in likelihood and severity) to the project and the services it is to provide over its lifespan.**

**3.2 Describe all of the risk mitigation measures that will be taken to improve the climate resiliency of your project.**

**3.3 Please list all of the climate change data and tools that were used to determine the risks to your project.**

### **4.0 Climate Objectives**

**4.1 Does your community / municipality have a Climate Action Plan and if yes, does your project align with this plan?**



## 5.0 Attestation

I, the undersigned, as authorized by my organization, confirm the statements above are true and accurate, and attest that:

- opportunities to quantify and minimize GHG emissions during the construction and operation of the project will be considered in the planning, design and development/ implementation of this project to the extent possible;
- and, climate change risks and adaptation and resiliency measures will be considered in the siting/location, design/build, and planned operation and maintenance of this project to the extent possible and reflecting the project's cost, criticality and vulnerability.

Infrastructure Canada may follow up on the results of the Climate Lens to confirm the required information or to request further detail. Consequently, applicants should retain all the information used to complete the Climate Lens.

Signature of person responsible for completing the Climate Lens:

X

Date:

Name:  
Position:  
Organization:  
Contact Number:  
Email:

## ANNEX B – Average P/T Grid Electricity Emission Intensities (tonnes/MWh)\*

| Region                | 2020  | 2021  | 2022  | 2023  | 2024  | 2025  | 2026  | 2027  | 2028  | 2029  | 2030  | 2031  | 2032  | 2033  | 2034  | 2035  |
|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Alberta               | 0.517 | 0.446 | 0.357 | 0.250 | 0.232 | 0.211 | 0.225 | 0.223 | 0.217 | 0.208 | 0.207 | 0.201 | 0.204 | 0.203 | 0.203 | 0.204 |
| British Columbia      | 0.004 | 0.002 | 0.003 | 0.003 | 0.004 | 0.002 | 0.002 | 0.002 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| Manitoba              | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| New Brunswick         | 0.276 | 0.259 | 0.269 | 0.268 | 0.275 | 0.273 | 0.274 | 0.272 | 0.258 | 0.252 | 0.124 | 0.116 | 0.124 | 0.113 | 0.123 | 0.114 |
| Newfoundland          | 0.091 | 0.068 | 0.012 | 0.012 | 0.012 | 0.011 | 0.011 | 0.011 | 0.011 | 0.010 | 0.010 | 0.010 | 0.011 | 0.010 | 0.010 | 0.009 |
| Northwest Territories | 0.058 | 0.067 | 0.062 | 0.051 | 0.017 | 0.008 | 0.008 | 0.008 | 0.010 | 0.012 | 0.014 | 0.016 | 0.020 | 0.014 | 0.013 | 0.009 |
| Nova Scotia           | 0.634 | 0.562 | 0.458 | 0.457 | 0.463 | 0.464 | 0.417 | 0.401 | 0.384 | 0.361 | 0.118 | 0.116 | 0.112 | 0.109 | 0.105 | 0.101 |
| Nunavut               | 0.747 | 0.747 | 0.744 | 0.712 | 0.635 | 0.498 | 0.480 | 0.469 | 0.470 | 0.455 | 0.457 | 0.442 | 0.435 | 0.447 | 0.454 | 0.458 |
| Ontario               | 0.034 | 0.044 | 0.067 | 0.065 | 0.066 | 0.077 | 0.093 | 0.081 | 0.067 | 0.064 | 0.062 | 0.060 | 0.058 | 0.041 | 0.035 | 0.030 |
| Prince Edward Island  | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| Quebec                | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| Saskatchewan          | 0.410 | 0.366 | 0.299 | 0.306 | 0.252 | 0.249 | 0.253 | 0.221 | 0.173 | 0.167 | 0.163 | 0.157 | 0.146 | 0.142 | 0.137 | 0.133 |
| Yukon Territory       | 0.045 | 0.121 | 0.068 | 0.077 | 0.086 | 0.089 | 0.099 | 0.074 | 0.046 | 0.029 | 0.018 | 0.014 | 0.018 | 0.023 | 0.032 | 0.041 |

| Region                | 2036  | 2037  | 2038  | 2039  | 2040  | 2041  | 2042  | 2043  | 2044  | 2045  | 2046  | 2047  | 2048  | 2049  | 2050  |
|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Alberta               | 0.206 | 0.207 | 0.209 | 0.210 | 0.212 | 0.213 | 0.215 | 0.216 | 0.217 | 0.219 | 0.220 | 0.221 | 0.221 | 0.221 | 0.222 |
| British Columbia      | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| Manitoba              | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| New Brunswick         | 0.124 | 0.111 | 0.118 | 0.114 | 0.129 | 0.129 | 0.120 | 0.121 | 0.122 | 0.124 | 0.125 | 0.126 | 0.128 | 0.130 | 0.131 |
| Newfoundland          | 0.009 | 0.009 | 0.009 | 0.009 | 0.008 | 0.008 | 0.008 | 0.008 | 0.008 | 0.008 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 |
| Northwest Territories | 0.008 | 0.006 | 0.006 | 0.006 | 0.008 | 0.025 | 0.026 | 0.031 | 0.020 | 0.018 | 0.016 | 0.017 | 0.019 | 0.020 | 0.022 |
| Nova Scotia           | 0.094 | 0.088 | 0.088 | 0.086 | 0.084 | 0.082 | 0.081 | 0.079 | 0.076 | 0.074 | 0.074 | 0.074 | 0.074 | 0.074 | 0.073 |
| Nunavut               | 0.470 | 0.482 | 0.488 | 0.488 | 0.501 | 0.505 | 0.515 | 0.523 | 0.525 | 0.529 | 0.535 | 0.544 | 0.547 | 0.556 | 0.561 |
| Ontario               | 0.024 | 0.021 | 0.019 | 0.017 | 0.016 | 0.015 | 0.015 | 0.015 | 0.014 | 0.013 | 0.011 | 0.009 | 0.009 | 0.011 | 0.013 |
| Prince Edward Island  | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| Quebec                | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| Saskatchewan          | 0.130 | 0.126 | 0.123 | 0.121 | 0.117 | 0.115 | 0.112 | 0.108 | 0.105 | 0.098 | 0.095 | 0.092 | 0.089 | 0.085 | 0.082 |
| Yukon Territory       | 0.054 | 0.067 | 0.052 | 0.039 | 0.027 | 0.019 | 0.013 | 0.017 | 0.020 | 0.026 | 0.033 | 0.042 | 0.050 | 0.034 | 0.022 |

Notes:

1. Grid Emissions intensity is defined as: (utility generation emissions) + (industrial net sales to grid by sector) x (industrial electricity generation emissions factor) *divided by* electricity consumption from the grid.
2. For alternative emission intensities from B.C. electricity, consult the provincial emission intensities found [here](#).

Source: ECCC's Greenhouse Gas Emissions Projections.

Link: [Canada's Greenhouse Gas Emissions Projections - Environment and Climate Change Canada Data](#).

Last Modified: June 2022.

## ANNEX C – Project-Specific GHG Baseline & Project Scenarios

The following provides an overview of various baseline and project scenarios acceptable under the Climate Lens.

| Project Type                                                                                   | Baseline                                                                                                                                                                                                                                 | Project                                                                                                                                                |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>New Building – Recreation or Sports Complex, Community Centre, Library, Housing Complex</b> | Facility designed and operated according to minimum building codes in province/municipality.                                                                                                                                             | Facility designed and operated with GHG measures above and beyond standard codes, such as use of solar energy, LEED design, low-carbon materials, etc. |
|                                                                                                | For facilities that are not being designed or operated with any additional GHG mitigation measures, the project = baseline and no operational GHG reductions will result from the project.                                               |                                                                                                                                                        |
| <b>Building/Facility Retrofits</b>                                                             | If no changes to building use or occupancy is expected, a historical 3 year average of all operational sources can be used. If changes are expected to use or occupancy, operational sources must be estimated to reflect these changes. | Facility retrofitted to increase energy efficiency and/or minimize fugitive emissions and/or installation of renewable energy.                         |
| <b>Facility or rural community installing Renewable Energy (Solar/Wind)</b>                    | Energy source that was being used previously or would be used in absence of the renewable energy source (e.g., diesel or natural gas).                                                                                                   | Renewable Energy generation (considered zero-emitting).                                                                                                |
| <b>Renewable Energy generation as part of “greening” the provincial grid</b>                   | Energy generation facility that would have been built instead of the renewable energy facility (On-the margin) (e.g., natural gas generation facility).                                                                                  | Renewable Energy generation (considered zero-emitting).                                                                                                |
| <b>Electrification of Industrial Facility</b>                                                  | Energy source that was being used previously or would be used in absence of the connection to the grid (e.g., diesel/natural gas).                                                                                                       | The provincial grid.                                                                                                                                   |
| <b>Fleet Replacement</b>                                                                       | Fuel that was being used or would be used instead of the new fuel source (e.g., diesel, gas).                                                                                                                                            | Clean energy source (e.g., electricity, hydrogen, biofuels).                                                                                           |
| <b>Large Transit (LRT/BRT)</b>                                                                 | The continuation of personal vehicle use without the new transit system.                                                                                                                                                                 | The transition (modal shifting) from personal vehicle use to public transit, which is considered less fuel - intensive.                                |
|                                                                                                | If the new transit system includes the construction of stations or storage facilities, the GHGs from these facilities should be quantified separately according to the new buildings baseline and project scenarios.                     |                                                                                                                                                        |

| Project Type                                                         | Baseline                                                                                                                                                                                                                                                                    | Project                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Active Transit (walking or biking paths)</b>                      | The continuation of personal vehicle use without the new active transit system.                                                                                                                                                                                             | The transition (modal shifting) from personal vehicle use to active transit, which is considered zero-emitting.                                                                                                                                     |
| <b>New Wastewater/Drinking Water Facilities</b>                      | Facility built to minimum building codes and wastewater/drinking water standards in province/municipality.                                                                                                                                                                  | Facility built with GHG measures beyond building codes & standards, such as implementation of energy efficient equipment or renewable energy or equipment/ processes to minimize fugitive or process emissions (CH <sub>4</sub> /N <sub>2</sub> O). |
|                                                                      | For infrastructure that is not being designed or operated with any additional GHG mitigation measures, the project = baseline and no operational GHG reductions will result from the project.                                                                               |                                                                                                                                                                                                                                                     |
| <b>Wastewater/Drinking Water Facility Retrofits</b>                  | If no changes to facility use or capacity (volume of water or wastewater treated) is expected, a historical 3 year average of all operational sources can be used. If changes are expected to facility use or capacity, sources must be estimated to reflect these changes. | Facility retrofitted to increase energy efficiency and/or minimize fugitive or process emissions (CH <sub>4</sub> /N <sub>2</sub> O) and/or installation of renewable energy.                                                                       |
| <b>Critical Infrastructure (Roads, Bridges, Culverts, Broadband)</b> | Infrastructure designed and operated according to minimum building codes/ standard practices in province/municipality.                                                                                                                                                      | Infrastructure designed and operated with GHG measures beyond standard codes. For e.g., use of low-carbon materials or vehicles/equipment using clean fuels.                                                                                        |
|                                                                      | For infrastructure that is not being designed or operated with any additional GHG mitigation measures, the project = baseline and no operational GHG reductions will result from the project.                                                                               |                                                                                                                                                                                                                                                     |
| <b>Disaster Mitigation and Adaptation projects</b>                   | Infrastructure designed and operated according to minimum building codes/ standard practices.                                                                                                                                                                               | Infrastructure designed and operated with GHG measures beyond standard codes. For e.g., use of low-carbon materials or vehicles/equipment using clean fuels.                                                                                        |
|                                                                      | For infrastructure that is not being designed or operated with any additional GHG mitigation measures, the project = baseline and no operational GHG reductions will result from the project.                                                                               |                                                                                                                                                                                                                                                     |
| <b>Landfills</b>                                                     | Facility meeting minimum standard regulations.                                                                                                                                                                                                                              | Facility flaring (where not mandated by regulations) or capturing methane for biogas production.                                                                                                                                                    |